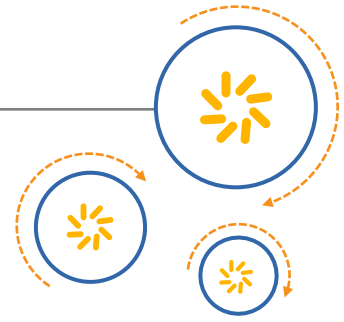




Qualcomm Technologies, Inc.



# GPS Quality, 19.2 MHz, Crystal, and TH+Xtal

## Mini Specification

LM80-P0436-39 Rev. B

January 9, 2018

© 2015, 2018 Qualcomm Technologies, Inc. All rights reserved.

MSM and Qualcomm Snapdragon are products of Qualcomm Technologies, Inc. Other Qualcomm products referenced herein are products of Qualcomm Technologies, Inc. or its other subsidiaries.

DragonBoard, MSM, Qualcomm and Snapdragon are trademarks of Qualcomm Incorporated, registered in the United States and other countries. All Qualcomm Incorporated trademarks are used with permission. Other product and brand names may be trademarks or registered trademarks of their respective owners.

This technical data may be subject to U.S. and international export, re-export, or transfer ("export") laws. Diversion contrary to U.S. and international law is strictly prohibited.

Use of this document is subject to the license set forth in Exhibit 1.

Qualcomm Technologies, Inc.  
5775 Morehouse Drive  
San Diego, CA 92121  
U.S.A.

LM80-P0436-39 Rev. A

## Revision history

| Revision | Date         | Description       |
|----------|--------------|-------------------|
| A        | August 2015  | Initial release   |
| B        | January 2018 | Added Appendix 7. |

# Contents

---

- 1 Purpose ..... 5**
- 2 19.2 MHz crystal specifications..... 6**
- 3 GPS Quality Specifications ..... 8**
- 4 Thermistor Specifications .....14**
- A How to Find the Curve-Fitting Parameters .....15**
  - A.1 FT data .....15
  - A.2 C0 15
  - A.3 C1 15
  - A.4 C3 and C2 .....15
  - A.5 Coefficient range .....15
  - A.6 Comments .....16
  - A.7 Example part numbers .....16
- B Exhibit 1 .....17**

Figures

Figure 3-1 Crystal perturbation specification 3 (DLD testing requirements) .....9

Figure 3-2 Measurement to locate resonances or micro-jumps .....11

Figure 3-3 Measurement to locate small orbit hysteresis 1 .....12

Figure 3-4 Measurement to locate hysteresis 2 .....13

Figure 4-1 TH+Xtal connection diagram.....14

Tables

Table 2-1 19.2 MHz crystal specifications .....6

Table 3-1 Crystal perturbation specification 1 (residual frequency stability slope) .....8

Table 3-2 Crystal perturbation specification 2 (small orbit hysteresis 2) .....8

Table 3-3 Specification for DLD .....9

Table 4-1 Thermistor specifications table .....14

# 1 Purpose

---

The operation, repeatability, and accuracy of a GPS receiver subsystem over time and temperature changes is highly dependent upon the quality of the reference crystal. This document shows how to test and confirm that a crystal meets the requirements of the GPS subsystem.

## 2 19.2 MHz crystal specifications

The crystal is a standard 4-pin crystal footprint with the crystal connected between pins 1 and 3.

**Table 2-1 19.2 MHz crystal specifications**

| Parameter                                                                                                                           | Min    | Nom                   | Max   | Units                                   | Notes                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------|-------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operating frequency                                                                                                                 |        | 19.2                  |       | MHz                                     |                                                                                                                                                                                                          |
| Mode of vibration                                                                                                                   |        | AT-cut<br>fundamental |       |                                         |                                                                                                                                                                                                          |
| Initial frequency tolerance                                                                                                         |        |                       | ±10   | ppm                                     |                                                                                                                                                                                                          |
| Tolerance over temperature                                                                                                          |        |                       | ±12   | ppm                                     | -30 to +85°C; above 85°C tolerance over temperature bound by third-order coefficient range in Section A.5.                                                                                               |
| Aging                                                                                                                               |        |                       | ±0.7  | ppm/year                                |                                                                                                                                                                                                          |
| Frequency drift after reflow                                                                                                        |        |                       | ±2    | ppm                                     | After two reflows                                                                                                                                                                                        |
| Operating temperature                                                                                                               | -30    |                       | +105  | °C                                      |                                                                                                                                                                                                          |
| Storage temperature                                                                                                                 | -40    |                       | +105  | °C                                      |                                                                                                                                                                                                          |
| Equivalent series resistance                                                                                                        |        |                       | 80    | Ω                                       | New for 2520 crystals                                                                                                                                                                                    |
| Quality factor (Q)                                                                                                                  | 75,000 |                       |       |                                         | The minimum Q value calculated from ESR and L is smaller than this specification.                                                                                                                        |
| Spurious mode series resistance                                                                                                     | 1100   |                       |       | Ω                                       | ±1 MHz                                                                                                                                                                                                   |
| Motional capacitance                                                                                                                | 1.80   |                       | 3.10  | fF                                      | New for 2520 crystals                                                                                                                                                                                    |
| Shunt capacitance                                                                                                                   | 0.3    |                       | 1.3   | pF                                      |                                                                                                                                                                                                          |
| Load capacitance                                                                                                                    |        | 7                     |       | pF                                      | The load capacitance is measured according to IEC Standard #60444-7.                                                                                                                                     |
| First-order curve fitting parameter (See Appendix A.) Note: See the attached Curve Fit Calculation Table to view the calculations.  | -0.40  |                       | -0.10 | ppm/°C                                  | The curve fitting parameter is obtained from the crystal curve fitting algorithm (Appendix A, calculation order C0, C1, C3, and C2) using the temperature inflection point at $t_0 = 30^\circ\text{C}$ . |
| Second-order curve fitting parameter (See Appendix A.) Note: See the attached Curve Fit Calculation Table to view the calculations. | -4.5   | 0                     | 4.5   | $\times 10^{-4}$<br>ppm/°C <sup>2</sup> | The curve fitting parameter is obtained from the crystal curve fitting algorithm (Appendix A, calculation order C0, C1, C3, and C2) using the temperature inflection point at $t_0 = 30^\circ\text{C}$ . |

| Parameter                                                                                                                             | Min | Nom       | Max  | Units                                   | Notes                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Third-order curve fitting parameter (See Appendix A.)<br>Note: See the attached Curve Fit Calculation Table to view the calculations. | 8.5 | 10        | 11.5 | $\times 10^{-5}$<br>ppm/°C <sup>3</sup> | The curve fitting parameter is obtained from the crystal curve fitting algorithm (Appendix A, calculation order C0, C1, C3, and C2) using the temperature inflection point at $t_0 = 30^\circ\text{C}$ . |
| Drive level                                                                                                                           | 10  |           | 100  | μW                                      |                                                                                                                                                                                                          |
| Insulation resistance                                                                                                                 | 500 |           |      | MΩ                                      |                                                                                                                                                                                                          |
| Package                                                                                                                               |     | 2.5 × 2.0 |      | mm <sup>2</sup>                         |                                                                                                                                                                                                          |

# 3 GPS Quality Specifications

**Table 3-1 Crystal perturbation specification 1 (residual frequency stability slope)**

| Item                                                                                      | Condition         | Specification | Unit   |
|-------------------------------------------------------------------------------------------|-------------------|---------------|--------|
| Residual frequency stability slope<br>(residual = difference from fifth-order curve fit)* | Ta = -30 to +85°C | ±50 (max)     | ppb/°C |
| 5°C small orbit hysteresis 1*                                                             | Ta = -30 to +85°C | ±50 (max)     | ppb/°C |

\*Must meet the 1A and 1B conditions.

- Condition 1A – Test condition (continuous temperature rate change of ~1.0°C/min)
  - Measure FT points every 1°C, heating up from -30 to +85°C, subtract a fifth-order polynomial best fit and then calculate the slope of the residual.
  - The residual slope should be within ±50 ppb/°C.
- Condition 1B – Hysteresis 1 test condition (continuous temperature rate change of ~1.0°C/min)
  - Measure FT points every 0.5°C while cycling temperature over a 5°C small temperature orbit. An example 5°C small orbit temperature cycle is +30°C to +35°C to +30°C.
  - During every individual heating/cooling cycle, there should be 11 points. Discard the first point of each heating and cooling cycle. This leaves 10 points for each heating and cooling cycle. Subtract the fifth-order polynomial best fit from 1A for each of the 10 points, and then calculate the slope of the residual for each of these heating and cooling 10 point curves.
  - The residual slope should be within ±50 ppb/°C.

**Table 3-2 Crystal perturbation specification 2 (small orbit hysteresis 2)**

| Item                           | Condition         | Specification   | Unit      |
|--------------------------------|-------------------|-----------------|-----------|
| 5°C small orbit hysteresis 2** | Ta = -30 to +85°C | 100 (magnitude) | ppb pk-pk |

\*\*Must meet condition 2

- Condition 2 – Hysteresis 2 test condition (continuous temperature rate change of ~1.0°C/min):
  - Measure FT points every 0.5°C while cycling temperature over a 5°C small temperature orbit; an example 5°C small orbit temperature cycle is +30°C to +35°C to +30°C.
  - During every individual heating/cooling cycle there should be 11 points; discard the first and last point of each heating and cooling cycle, which results in 9 temperature points. Calculate the average measured peak-to-peak frequency difference for these 9 temperature points.
  - The average difference is the magnitude of the small orbit hysteresis 2.

**Table 3-3 Specification for DLD**

|                        | Item | Max – min | Repeatability | Condition                    |
|------------------------|------|-----------|---------------|------------------------------|
| Drive level dependency | Freq | < 3 ppm   | < 0.7 ppm     | 0.01 uW to 100 uW to 0.01 uW |
|                        | ESR  | < 20%     | < 10%         |                              |

**Note:** Number of points: 15 points up and 15 points down = 29 total data points

Max - min: Difference between maximum and minimum in two-way measurement.

In case of ESR, the change rate is  $(\text{max} - \text{min})/\text{min} < 20\%$ .

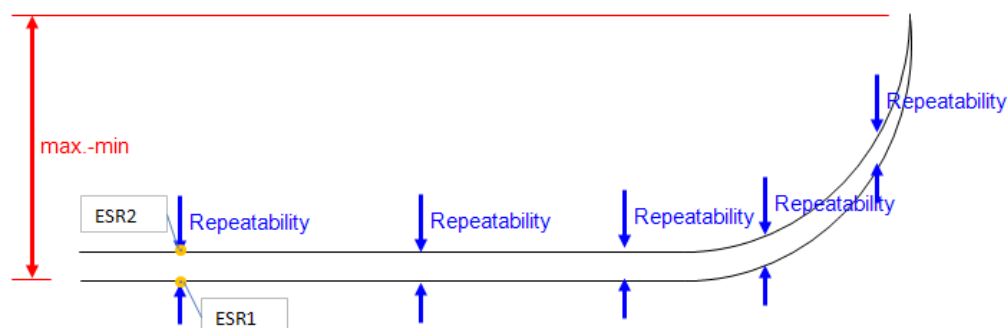
Repeatability: Repeatability of two-way measurement in the above condition.

In case of ESR, the change rate is  $(\text{ESR2} - \text{ESR1})/\text{ESR1} < 10\%$ .

ESR1: First measurement on each drive levels

ESR2: Second measurement on each drive levels

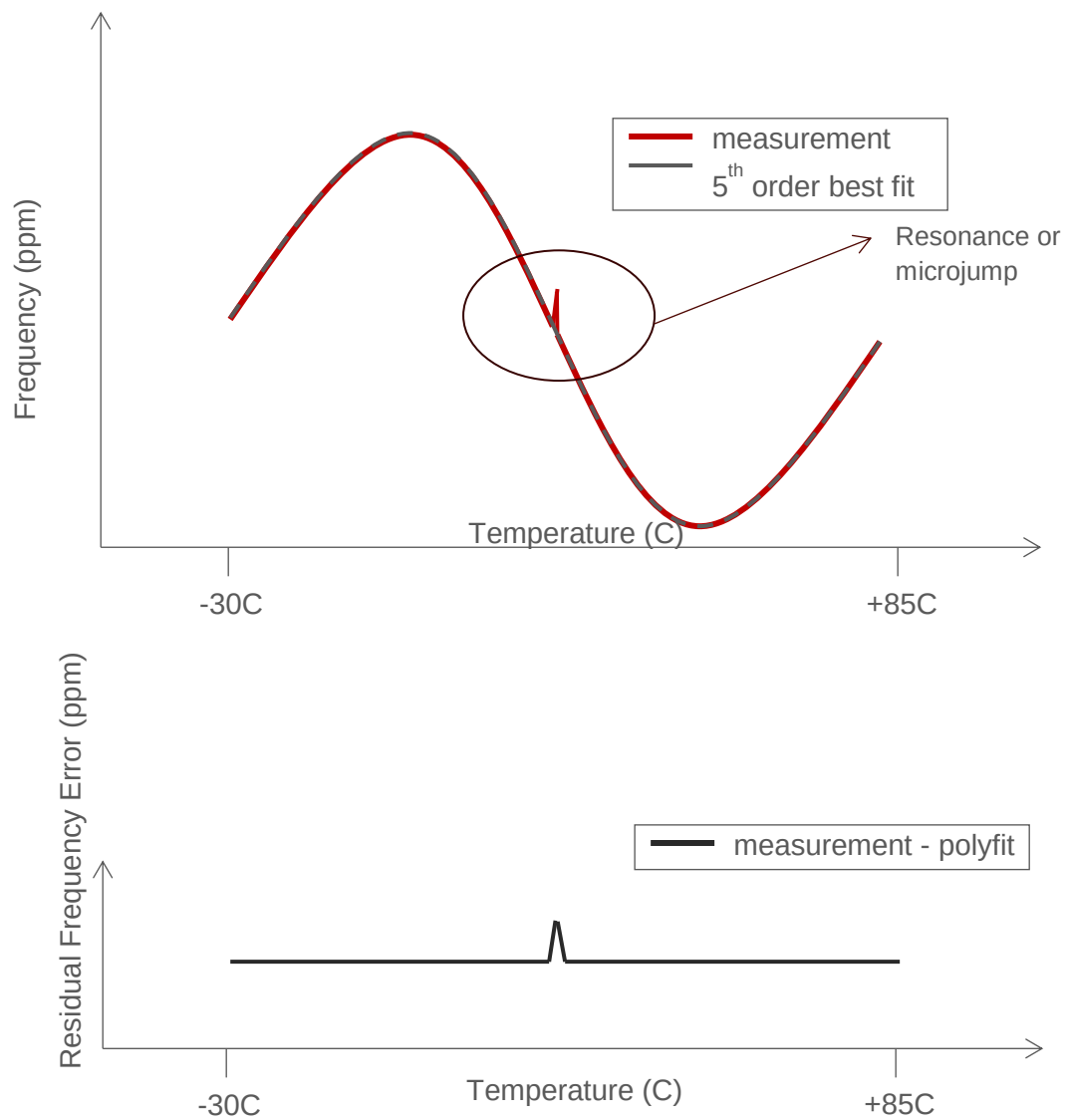
(Example) How to specify each parameter

**Figure 3-1 Crystal perturbation specification 3 (DLD testing requirements)**

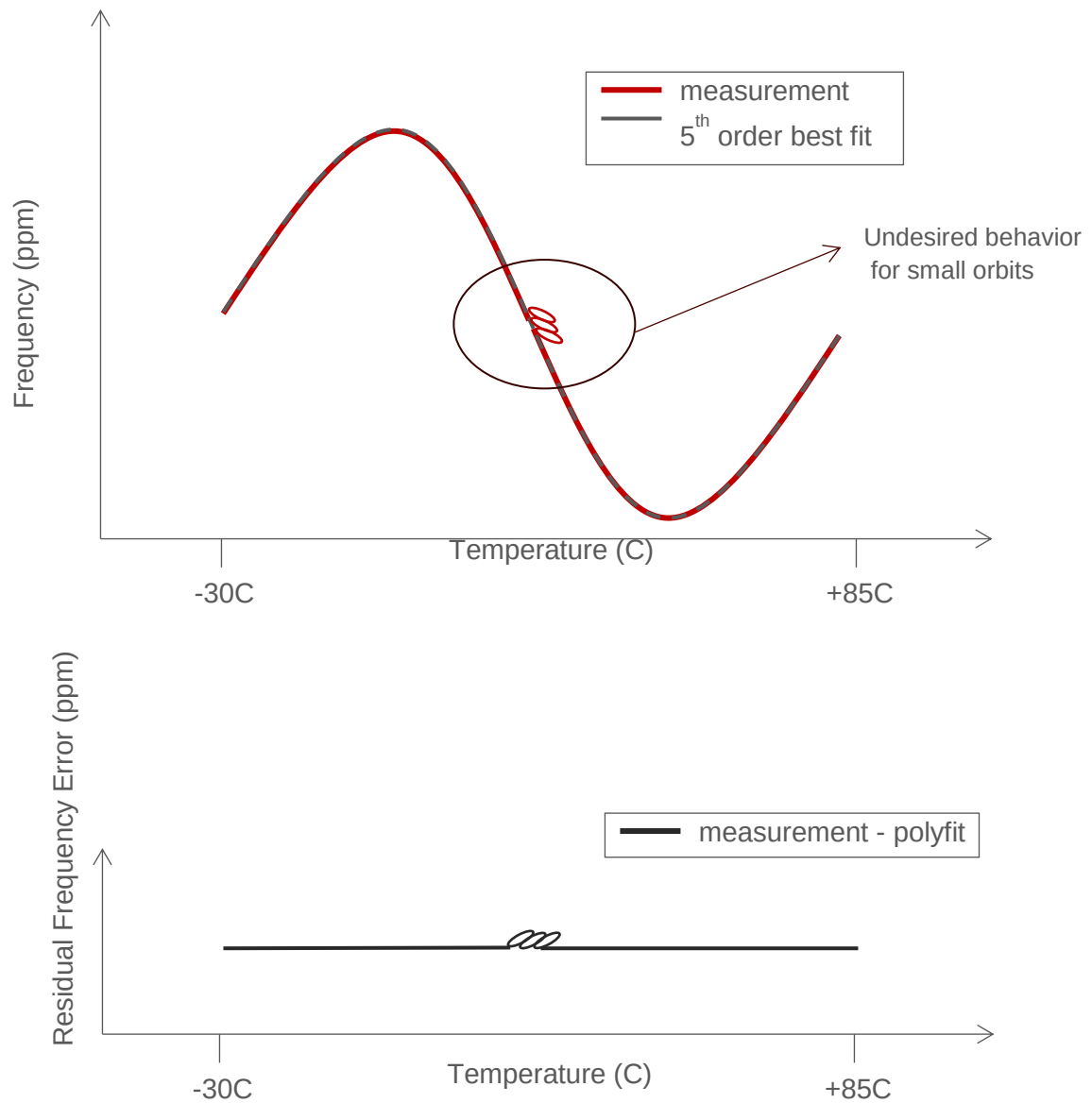
The key points are as follows:

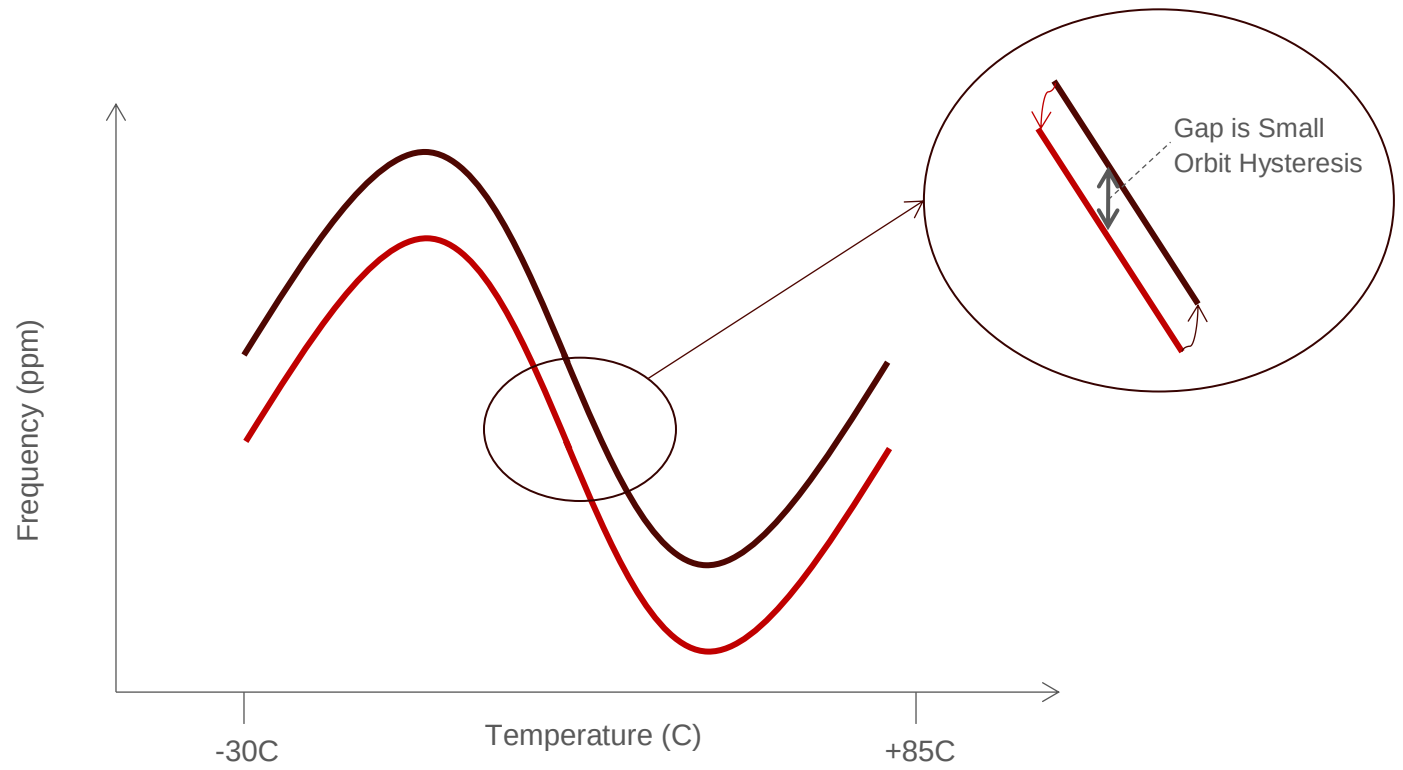
1. Testing:
  - ❑ DLD testing should be done on every device to detect fabrication issues such as contamination, particles of dust, etc., and to verify proper crystal functionality.
  - ❑ Specifications 1A, 1B, and 2 process control confirmation testing is recommended to be done at a minimum on a sampling basis for every LOT.
2. The purpose of the specifications:
  - ❑ Specification 1A:
    - Verify that the FT curve can fit with a fifth-order polynomial.
    - Verify that there are no package resonances and no micro-jumps that exceed the residual frequency stability slope specification.
    - See [Figure 3-2](#).
  - ❑ Specification 1B:
    - Verify that the small orbit hysteresis 1 is controlled to less than 50 ppb/°C residual slope in the crystal (departure from FT curve over small orbits).
    - See [Figure 3-3](#).

- ❑ Specification 2:
  - Verify that hysteresis 2 (the gap between the cooling and heating FT curves) over small orbits is of a reasonable value.
  - These are the rapid shifts that occur when changing from cooling-to-heating or heating-to-cooling.
  - See [Figure 3-4](#).
- 3. Measurement technique for specifications 1 and 2:
  - ❑ Standard thermocouple should not be used since it has a 0.1°C temperature resolution and therefore can fail a 50 ppb/°C test due to measurement quantization (i.e., which is not part of the real crystal FT performance curve).
  - ❑ A preferable technique to measure temperature is to use the internal thermistor (or an external thermistor located near crystal on the PCB board). As an example, voltage measurement is a feature of the 34970A Data Acquisition Unit from Agilent. Frequency can be measured via the 53131A\_132A 225 MHz Agilent frequency counter.
  - ❑ A continuous temperature rate change of ~1.0°C/min is recommended for all tests.



**Figure 3-2 Measurement to locate resonances or micro-jumps**

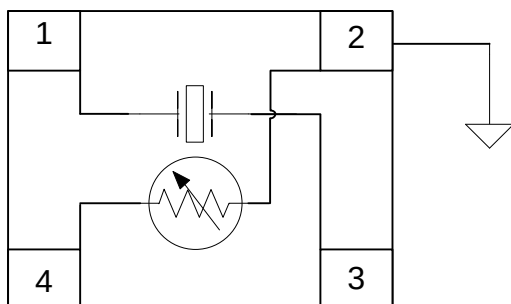
**Figure 3-3 Measurement to locate small orbit hysteresis 1**

**Figure 3-4 Measurement to locate hysteresis 2**

## 4 Thermistor Specifications

The thermistor can be discrete on the CCA next to the crystal or collocated within the crystal package.

If the thermistor is in the crystal package, the connection of the crystal and thermistor should be as shown in [Figure 4-1](#). Pin 1 and pin 3 are the two terminals of the crystal; pin 2 is the ground and provides ground reference to the lid of the package. The thermistor is connected between pin 4 and terminated on pin 2. This integrated thermistor and crystal device is referred to as TH+Xtal. The crystal must meet 100% of the crystal specifications in [Table 2-1](#), and the thermistor must meet 100% of the thermistor specifications in [Table 3-3](#).



**Figure 4-1 TH+Xtal connection diagram**

**Table 4-1 Thermistor specifications table**

| Parameter             | Min | Nom  | Max  | Units | Notes   |
|-----------------------|-----|------|------|-------|---------|
| Operating temperature | -30 |      | +105 | °C    |         |
| Storage temperature   | -40 |      | +105 | °C    |         |
| Resistance            |     | 100  |      | kΩ    | 25°C    |
| B-constant            |     | 4250 |      | K     | 25–50°C |
| Tolerance             |     |      | 1    | %     |         |

# A How to Find the Curve-Fitting Parameters

---

The FT curve of an AT-cut crystal can be modeled as a third-order polynomial.

$$f(t) = c_3(t - t_0)^3 + c_2(t - t_0)^2 + c_1(t - t_0) + c_0$$

C0, C1, C2, and C3 are coefficients that need to be defined. The C1 and C3 parameter of the crystals must fall into a certain predetermined range for the system to estimate the resonant frequency of the crystal effectively. In the following sections, the details of the coefficient calculations are presented, and a range is listed of acceptable C1 and C3 coefficients.

## A.1 FT data

The crystal vendors need to gather the FT data. The FT data should cover a temperature range from -30 to 105°C. The resolution should be 1°C (linear extrapolation can be used), and the frequency value should be in ppm and reflect the deviation from the frequency at 30°C.

## A.2 C0

C0 is the frequency offset at 30°C.

## A.3 C1

C1 is the average slope of the FT curve between 25°C and 35°C.

## A.4 C3 and C2

Refer to the *Curve Fit Calculation Table* for the C3 and C2 parameter calculations.

- To calculate C3, find the difference between the measured FT curve and the first-order estimation from the C1 parameter. Next, find the C3 term using the third-order least square fit method. Use 30°C as  $t_0$ .
- To calculate C2, remove the third-order and first-order terms first from the FT curve. Next, find the C2 term using the second-order least square fit method. Use 30°C as  $t_0$ .

## A.5 Coefficient range

- C0:  $\pm 10$  ppm
- C1: -0.10 to -0.40 ppm/°C
- C2:  $-4.5 \times 10^{-4}$  to  $4.5 \times 10^{-4}$  ppm/°C<sup>2</sup>
- C3:  $8.5 \times 10^{-5}$  to  $11.5 \times 10^{-5}$  ppm/°C<sup>3</sup>

## A.6 Comments

C3 needs to be calculated before C2.

**NOTE:** Open the embedded file attached, to see the Curve Fit Calculation Table, to view the calculations in an Excel file.

## A.7 Example part numbers

The following crystals were tested and found to work when this document was published. It is the responsibility of the system designer and manufacturer to verify and validate that the selected part is suitable for the intended uses. Crystal manufacturers will obsolete the parts from time to time, or make new parts available, this is not a comprehensive list.

| Vendor                                        | Part #               | Size/Package |
|-----------------------------------------------|----------------------|--------------|
| Epson China Co., Ltd                          | X1E000291000100      | 2520         |
| Epson China Co., Ltd                          | X1E0002910002        | 2520         |
| Epson China Co., Ltd                          | X1E0003010002        | 2520         |
| Epson China Co., Ltd                          | X1E0003610003        | 2016         |
| Epson China Co., Ltd                          | X1E0003710001        | 2016         |
| Shanghai Daishinku Int'l Trading Company Ltd. | 1MAA19200AAA         | 2520         |
| Shanghai Daishinku Int'l Trading Company Ltd. | 1MAA19200ACA         | 2520         |
| Shanghai Daishinku Int'l Trading Company Ltd. | 1RAE19200BAA         | 2016         |
| Shanghai Daishinku Int'l Trading Company Ltd. | 1RAE19200BCA         | 2016         |
| Shanghai Daishinku Int'l Trading Company Ltd. | 1ZH219200AB0D        | 2520         |
| Kyocera Corporation                           | CT2016DB19200C0FLHA1 | 2016         |
| Kyocera Corporation                           | CT2016DB19200C0FLHA9 | 2016         |
| Kyocera Corporation                           | CT2016KB19200C0FLHA1 | 2016         |
| Kyocera Corporation                           | CT2520DB19200C0FLHA1 | 2520         |
| Kyocera Corporation                           | CT2520DB19200C0FLHAF | 2520         |
| Kyocera Corporation                           | CX2520DB19200C0FLHA4 | 2520         |
| NDK ELECTRONICS SHANGHAI CO., LTD             | EXS00A-CS04340       | 2520         |
| NDK ELECTRONICS SHANGHAI CO., LTD             | EXS00A-CS05052       | 2520         |
| NDK ELECTRONICS SHANGHAI CO., LTD             | EXS00A-CS05053       | 2520         |
| NDK ELECTRONICS SHANGHAI CO., LTD             | EXS00A-CS05857       | 2016         |
| NDK ELECTRONICS SHANGHAI CO., LTD             | EXS00A-CS06709       | 2016         |
| Rakon                                         | 508050               | 2520         |
| Rakon                                         | 509064               | 2016         |
| TEW                                           | 120-17540A           | 2520         |
| TXC Corporation                               | 8Z19270004           | 2520         |
| TXC Corporation                               | OY19270003           | 2016         |
| TXC Corporation                               | OY19270004           | 2016         |
| TXC Corporation                               | OZ19200001           | 2520         |
| TXC Corporation                               | OZ19270001           | 2520         |

# B Exhibit 1

PLEASE READ THIS LICENSE AGREEMENT ("AGREEMENT") CAREFULLY. THIS AGREEMENT IS A BINDING LEGAL AGREEMENT ENTERED INTO BY AND BETWEEN YOU (OR IF YOU ARE ENTERING INTO THIS AGREEMENT ON BEHALF OF AN ENTITY, THEN THE ENTITY THAT YOU REPRESENT) AND QUALCOMM TECHNOLOGIES, INC. ("QTI" "WE" "OUR" OR "US"). THIS IS THE AGREEMENT THAT APPLIES TO YOUR USE OF THE DESIGNATED AND/OR ATTACHED DOCUMENTATION AND ANY UPDATES OR IMPROVEMENTS THEREOF (COLLECTIVELY, "MATERIALS"). BY USING OR COMPLETING THE INSTALLATION OF THE MATERIALS, YOU ARE ACCEPTING THIS AGREEMENT AND YOU AGREE TO BE BOUND BY ITS TERMS AND CONDITIONS. IF YOU DO NOT AGREE TO THESE TERMS, QTI IS UNWILLING TO AND DOES NOT LICENSE THE MATERIALS TO YOU. IF YOU DO NOT AGREE TO THESE TERMS YOU MUST DISCONTINUE AND YOU MAY NOT USE THE MATERIALS OR RETAIN ANY COPIES OF THE MATERIALS. ANY USE OR POSSESSION OF THE MATERIALS BY YOU IS SUBJECT TO THE TERMS AND CONDITIONS SET FORTH IN THIS AGREEMENT.

1.1 **License.** Subject to the terms and conditions of this Agreement, including, without limitation, the restrictions, conditions, limitations and exclusions set forth in this Agreement, Qualcomm Technologies, Inc. ("QTI") hereby grants to you a nonexclusive, limited license under QTI's copyrights to use the attached Materials; and to reproduce and redistribute a reasonable number of copies of the Materials. You may not use Qualcomm Technologies or its affiliates or subsidiaries name, logo or trademarks; and copyright, trademark, patent and any other notices that appear on the Materials may not be removed or obscured. QTI shall be free to use suggestions, feedback or other information received from You, without obligation of any kind to You. QTI may immediately terminate this Agreement upon your breach. Upon termination of this Agreement, Sections 1.2-4 shall survive.

1.2 **Indemnification.** You agree to indemnify and hold harmless QTI and its officers, directors, employees and successors and assigns against any and all third party claims, demands, causes of action, losses, liabilities, damages, costs and expenses, incurred by QTI (including but not limited to costs of defense, investigation and reasonable attorney's fees) arising out of, resulting from or related to: (i) any breach of this Agreement by You; and (ii) your acts, omissions, products and services. If requested by QTI, You agree to defend QTI in connection with any third party claims, demands, or causes of action resulting from, arising out of or in connection with any of the foregoing.

1.3 **Ownership.** QTI (or its licensors) shall retain title and all ownership rights in and to the Materials and all copies thereof, and nothing herein shall be deemed to grant any right to You under any of QTI's or its affiliates' patents. You shall not subject the Materials to any third party license terms (e.g., open source license terms). You shall not use the Materials for the purpose of identifying or providing evidence to support any potential patent infringement claim against QTI, its affiliates, or any of QTI's or QTI's affiliates' suppliers and/or direct or indirect customers. QTI hereby reserves all rights not expressly granted herein.

1.4 **WARRANTY DISCLAIMER.** YOU EXPRESSLY ACKNOWLEDGE AND AGREE THAT THE USE OF THE MATERIALS IS AT YOUR SOLE RISK. THE MATERIALS AND TECHNICAL SUPPORT, IF ANY, ARE PROVIDED "AS IS" AND WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESS OR IMPLIED. QTI ITS LICENSORS AND AFFILIATES MAKE NO WARRANTIES, EXPRESS OR IMPLIED, WITH RESPECT TO THE MATERIALS OR ANY OTHER INFORMATION OR DOCUMENTATION PROVIDED UNDER THIS AGREEMENT, INCLUDING BUT NOT LIMITED TO ANY WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE OR AGAINST INFRINGEMENT, OR ANY EXPRESS OR IMPLIED WARRANTY ARISING OUT OF TRADE USAGE OR OUT OF A COURSE OF DEALING OR COURSE OF PERFORMANCE. NOTHING CONTAINED IN THIS AGREEMENT SHALL BE CONSTRUED AS (I) A WARRANTY OR REPRESENTATION BY QTI, ITS LICENSORS OR AFFILIATES AS TO THE VALIDITY OR SCOPE OF ANY PATENT, COPYRIGHT OR OTHER INTELLECTUAL PROPERTY RIGHT OR (II) A WARRANTY OR REPRESENTATION BY QTI THAT ANY MANUFACTURE OR USE WILL BE FREE FROM INFRINGEMENT OF PATENTS, COPYRIGHTS OR OTHER INTELLECTUAL PROPERTY RIGHTS OF OTHERS, AND IT SHALL BE THE SOLE RESPONSIBILITY OF YOU TO MAKE SUCH DETERMINATION AS IS NECESSARY WITH RESPECT TO THE ACQUISITION OF LICENSES UNDER PATENTS AND OTHER INTELLECTUAL PROPERTY OF THIRD PARTIES.

1.5 **LIMITATION OF LIABILITY.** IN NO EVENT SHALL QTI, QTI'S AFFILIATES OR ITS LICENSORS BE LIABLE TO YOU FOR ANY INCIDENTAL, CONSEQUENTIAL OR SPECIAL DAMAGES, INCLUDING BUT NOT LIMITED TO ANY LOST PROFITS, LOST SAVINGS, OR OTHER INCIDENTAL DAMAGES, ARISING OUT OF THE USE OR INABILITY TO USE, OR THE DELIVERY OR FAILURE TO DELIVER, ANY OF THE MATERIALS, OR ANY BREACH OF ANY OBLIGATION UNDER THIS AGREEMENT, EVEN IF QTI HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. THE FOREGOING LIMITATION OF LIABILITY SHALL REMAIN IN FULL FORCE AND EFFECT REGARDLESS OF WHETHER YOUR REMEDIES HEREUNDER ARE DETERMINED TO HAVE FAILED OF THEIR ESSENTIAL PURPOSE. THE ENTIRE LIABILITY OF QTI, QTI'S AFFILIATES AND ITS LICENSORS, AND THE SOLE AND EXCLUSIVE REMEDY OF YOU, FOR ANY CLAIM OR CAUSE OF ACTION ARISING HEREUNDER (WHETHER IN CONTRACT, TORT, OR OTHERWISE) SHALL NOT EXCEED US\$10.

2. **COMPLIANCE WITH LAWS; APPLICABLE LAW.** You agree to comply with all applicable local, international and national laws and regulations and with U.S. Export Administration Regulations, as they apply to the subject matter of this Agreement. This Agreement is governed by the laws of the State of California, excluding California's choice of law rules.

3. **CONTRACTING PARTIES.** If the Materials are downloaded on any computer owned by a corporation or other legal entity, then this Agreement is formed by and between QTI and such entity. The individual accepting the terms of this Agreement represents and warrants to QTI that they have the authority to bind such entity to the terms and conditions of this Agreement.

4. **MISCELLANEOUS PROVISIONS.** This Agreement, together with all exhibits attached hereto, which are incorporated herein by this reference, constitutes the entire agreement between QTI and You and supersedes all prior negotiations, representations and agreements between the parties with respect to the subject matter hereof. No addition or modification of this Agreement shall be effective unless made in writing and signed by the respective representatives of QTI and You. The restrictions, limitations, exclusions and conditions set forth in this Agreement shall apply even if QTI or any of its affiliates becomes aware of or fails to act in a manner to address any violation or failure to comply therewith. You hereby acknowledge and agree that the restrictions, limitations, conditions and exclusions imposed in this Agreement on the rights granted in this Agreement are not a derogation of the benefits of such rights. You further acknowledges that, in the absence of such restrictions, limitations, conditions and exclusions, QTI would not have entered into this Agreement with You. Each party shall be responsible for and shall bear its own expenses in connection with this Agreement. If any of the provisions of this Agreement are determined to be invalid, illegal, or otherwise unenforceable, the remaining provisions shall remain in full force and effect. This Agreement is entered into solely in the English language, and if for any reason any other language version is prepared by any party, it shall be solely for convenience and the English version shall govern and control all aspects. If you are located in the province of Quebec, Canada, the following applies: The Parties hereby confirm they have requested this Agreement and all related documents be prepared in English.